J.P. MOLASSES, INC.

Production Planning Optimization

Presented to Cyril Morel, General Manager

Group 1-4 | EGRMGMT 580 – Decision Models

$149,178	$1,122,146	100%	7/7
Monthly Profit	Monthly Revenue	Customer Demand Met	Customers Served

Agenda

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Business context, objectives, and key constraints

02 **Optimization Model**

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03 **Recommended Supply Plan**

Optimal raw sugar allocation across both refineries

04 **Molasses Distribution**

Optimal customer assignment from each refinery

05 **Financial Results**

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06 **Sensitivity Analysis**

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The Challenge

Problem Overview

The Challenge

1. J.P. Molasses operates two refineries : Charleston (SC) and Atlanta (GA) , purchasing raw sugar from 8 suppliers across the southern US.
2. Refineries must supply molasses to 7 customer factories: 5 internal (no payment) and 2 external (Shorewood & St. Mary's, pay $36/K lbs market price).
3. Surplus molasses after customers are served is sold on the open market at $36/K lbs.
4. Current manual planning is slow, error-prone, and cannot evaluate alternative plans efficiently.
5. Objective: Maximize monthly profit by optimally routing raw sugar to refineries and molasses to customers.

8 Suppliers

Raw sugar sources, Southern US each with fixed price, freight cost, and monthly supply cap

2 Refineries

Must operate 50–100% of capacity. Charleston: $31/K lbs variable cost. Atlanta: $38/K lbs.

7 Customers

All demand must be 100% satisfied. Refineries cannot buy molasses externally to cover shortfalls.

Key Physical Limits

2.7% transport loss. Charleston sugar cap: 2,000 K lbs/mo. Charleston yield: 42.1% molasses, 47.9% sugar.

Optimization Model Structure

Decision Variables

› x(s,C) = K lbs of raw sugar from supplier s → Charleston [8 vars]
› x(s,A) = K lbs of raw sugar from supplier s → Atlanta [8 vars]
› y(c,C) = K lbs of molasses from Charleston → customer c [7 vars]
› y(c,A) = K lbs of molasses from Atlanta → customer c [7 vars]
› 30 continuous, non-negative decision variables total

Objective : Maximize Profit

› + Sugar revenue: Charleston @$200, Atlanta @$150 per K lbs
› + External molasses revenue: Shorewood & St. Mary's @$36/K lbs
› + Surplus molasses open market: @$36/K lbs
› − Raw sugar purchase cost (f.o.b. supplier price × quantity)
› − Freight: suppliers → refineries (per K lbs shipped)
› − Variable processing: $31/K lbs Charleston, $38/K lbs Atlanta
› − Fixed costs: $10,000 Charleston + $14,200 Atlanta per month
› − Freight: refineries → all 7 customers (per K lbs delivered)

Constraints

› Supply: Total bought ≤ available per supplier (8 constraints)
› Capacity Min: raw sugar processed ≥ 50% of each plant's max
› Capacity Max: raw sugar processed ≤ 100% of each plant's max
› Sugar limit: Charleston purified sugar ≤ 2,000 K lbs/month
› Molasses out: shipped ≤ 0.973 × yield × raw sugar in (per plant)
› Demand: total delivered = required for each customer (7 equalities)
› Non-negativity: all quantities ≥ 0
› Solver: Simplex LP, optimal solution in 0.2 seconds

Optimal Raw Sugar Allocation — Supplier Routing

Supplier	Available (K lbs)	→ Charleston (K lbs)	→ Atlanta (K lbs)	All Used?	Routing Decision
Spectra	1,000	0	1,000	Yes ✓	Atlanta only , lower combined cost
Larkin	1,583	1,583	0	Yes ✓	Charleston only , lowest freight to CHS
Portia	2,140	0	2,140	Yes ✓	Atlanta only , Portia freight to CHS too high ($19.6)
Omega	1,370	708	662	Yes ✓	Split , optimal boundary; both plants compete for Omega
Thistle	2,000	2,000	0	Yes ✓	Charleston only , favorable freight ($4.2)
Milligan	1,850	0	1,850	Yes ✓	Atlanta only , higher CHS freight ($7.65 vs $11)
Harvester	1,260	0	1,260	Yes ✓	Atlanta only , Harvester freight to ATL very low ($4.8)
Coleridge	1,700	0	1,700	Yes ✓	Atlanta only , CHS freight too high ($16.3)
TOTAL	**12,903**	**4,291**	**8,612**	**8/8 ✓**	**All suppliers fully utilized**

All 8 suppliers fully exhausted. Omega is the only split supplier — Solver places it on the cost boundary between both plants. Charleston processes 4,175 K lbs post-loss (at 50% min floor). Atlanta processes 8,379 K lbs (95.4% of max capacity).

Optimal Molasses Distribution — Customer Routing

Customer	Required (K lbs)	From Charleston	From Atlanta	Type	Why This Routing
Quentin	480	480	0	Internal	CHS freight $26.0 vs ATL $30.1 : saves $1,968/mo
Ashberry	850	0	850	Internal	ATL freight $31.7 vs CHS $51.7 : saves $17,000/mo
Wilksboro	640	0	640	Internal	ATL freight $7.3 vs CHS $16.6 : saves $5,952/mo
Casey	575	575	0	Internal	CHS freight $16.2 vs ATL $21.5 : saves $3,048/mo
Walsh	970	0	970	Internal	ATL freight $13.2 vs CHS $24.5 : saves $10,961/mo
Shorewood	107	107	0	External ★	CHS freight $26.3 vs ATL $28.0 : external; generates revenue
St. Mary's	80	80	0	External ★	CHS freight $21.3 vs ATL $46.2 : CHS far cheaper
TOTAL	**3,702**	**1,242**	**2,460**	**7/7 ✓**	**All customers 100% served: zero shortfall**

Surplus Molasses Sold on Open Market (@$36/K lbs): Charleston surplus → 516 K lbs = $18,576/mo | Atlanta surplus → 573 K lbs = $20,632/mo | Combined: $39,208/mo

Financial Results — Monthly P&L

COSTS

Raw Sugar Purchase (f.o.b.)	**$312,478**
Freight: Suppliers → Refineries	**$105,281**
Variable Processing, Charleston	**$133,028**
Variable Processing , Atlanta	**$327,247**
Fixed Operating (both plants)	**$24,200**
Freight: Refineries → Customers	**$70,734**
TOTAL COSTS	**$972,968**

REVENUES

Charleston Sugar Sales (@$200/K lbs)	**$400,000**
Atlanta Sugar Sales (@$150/K lbs)	**$676,206**
External Customers (Shorewood + St. Mary's)	**$6,732**
Surplus Molasses, Open Market	**$39,208**
TOTAL REVENUE	**$1,122,146**

MONTHLY PROFIT

$149,178

Note: Atlanta variable processing ($327K) is the single largest cost item — Atlanta runs nearly 2.4× Charleston's throughput but at $7/K lbs higher unit cost.

Sensitivity Analysis — Real Solver Numbers

Binding Constraints — Shadow Prices from Solver

Charleston Sugar Storage Cap (2,000 K lbs)

Shadow Price: $65.14 / K lbs

Currently binding at exactly 2,000 K lbs of purified sugar.
For every additional 1 K lbs of storage capacity, monthly profit increases by $65.14.
This holds for up to +308 K lbs of additional capacity (range from Sensitivity Report).
Decision: evaluate if storage expansion CAPEX < $65.14 × incremental K lbs.

All 8 Supplier Supply Caps — Fully Utilized

Shadow Price: $15.00–$25.10 / K lbs (varies by supplier)

Spectra: $17.40 | Larkin: $17.10 | Portia: $16.20 | Omega: $19.30
Milligan: $18.90 | Thistle: $18.20 | Harvester: $25.10 | Coleridge: $18.70
Harvester has highest shadow price ($25.10) → most valuable to source more from.
All shadow prices represent the profit gain per additional K lbs of supply.

All 7 Customer Demand Constraints (equality)

Shadow Price (cost): –$21.30 to –$67.70 / K lbs

Quentin: –$62.00 | Casey: –$52.20 | Walsh: –$49.20 | Wilksboro: –$43.30
Shorewood: –$26.30 | St. Mary's: –$21.30 | Ashberry: –$67.70
Negative because serving customers is a cost (no revenue for internal ones).
Ashberry is most costly to serve ($67.70/K lbs) due to high Atlanta freight.

Non-Binding & Key Opportunities

Atlanta Max Capacity

Shadow price = 0

401 K lbs of unused capacity. No value in expanding Atlanta, it already has headroom.

Charleston Min Capacity

Shadow price = 0

Charleston already above the 50% floor (160 K lbs of slack). Not a binding concern.

Molasses Surplus Constraint

Shadow price = 0

Both plants have surplus: CHS 516 K lbs, ATL 573 K lbs, both sold on open market.

Reduced Cost Example

Harvester to Charleston: reduced cost = –$16.50/K lbs. This means Harvester-to-CHS freight is $16.50/K lbs too expensive before Solver would route any there. If Harvester's CHS freight dropped by $16.50 (from $14.70 → under ~$4), it would become worth routing.

Recommendations to Cyril Morel

01 Adopt the Optimal Plan Now

Implement: Larkin + Thistle + Omega(708) → Charleston; all others → Atlanta. Customers: Quentin, Casey, Shorewood, St. Mary's from Charleston; Ashberry, Wilksboro, Walsh from Atlanta. This delivers $149,178/month profit , use it as the baseline every year.

02 Expand Charleston Sugar Storage

The 2,000 K lbs storage cap is the single most profitable bottleneck. Shadow price = $65.14/K lbs of capacity added, valid up to +308 K lbs. If storage expansion costs < $65 × 308K = ~$20,000/mo amortized, it pays for itself. Get a capital quote.

03 Negotiate More Supply from Harvester

Harvester has the highest supplier shadow price: $25.10/K lbs. Currently fully utilized at 1,260 K lbs. Every additional K lbs Harvester can supply increases monthly profit by $25.10. Approach Harvester for volume expansion before others.

04 Use This Model as an Annual Planning Tool

Solver found the optimal solution in 0.2 seconds vs. days of manual planning. Refresh inputs annually, supplier prices, freight rates, customer volumes and re-run. Sensitivity report will instantly flag which constraints are newly binding.

CONCLUSION

Linear optimization delivers a clear, defensible, and profitable plan.

$149,178	$65.14/K lbs	$25.10/K lbs	0.2 seconds
Monthly profit — optimizer vs. guesswork	Value of relaxing Charleston storage cap	Value of each additional K lbs from Harvester	Solver runtime vs. days of manual planning

Immediate next step: get a capital cost estimate for Charleston sugar storage expansion, sensitivity analysis says it pays back at $65.14/K lbs of added capacity.